

Estimation of Change in Population of municipalities in PR

Alejandro M. Ouslan & Tatiana E. Crespo



[ESMA 6835]: Inferencia Bayesian

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Presentation Overview

- ① Introduction
- ② Methodology

- ③ Results
 - Densities
 - Confidence Interval
 - Credibility Interval
 - Distribution
- ④ Predictions
- ⑤ Conclusions

- Demographic change is an important topic for governments and policymakers.
- Understanding population characteristics is particularly important for regional governments.
- Anticipating population change helps guide infrastructure and public investment decisions.

- Data obtained from the American Community Survey (ACS) 5-year estimates U.S. Census Bureau
- Population estimates from 2011 to 2023 were used.
- Dataset includes all 78 municipalities in Puerto Rico.

The variables to consider for our simple regression model are:

- **Independent variable X :** the years for which we have population estimates for Puerto Rico
- **Dependent variable Y :** the total population for the 78 municipalities of Puerto Rico for each year

The simple linear regression model to be used is the following:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

where:

- Y_i = population in year i
- X_i = year
- β_0 = intercept
- β_1 = population change rate
- ϵ_i = random error

- A **classical linear regression** was performed with a significance level of $\alpha = 0.05$, from which we obtained a population change rate of -210.52 , indicating that the total population decreases each year.
- However, upon reviewing the p-value of this test (0.540), we realized that our findings are not statistically significant.
- Therefore, based on classical inference, there is insufficient evidence to determine that there is a drastic change in the population decrease over time in this model.

OLS Regression Results

Model Summary

Observations	1014
R-squared	0.000
Adj. R-squared	-0.001
F-statistic	0.3759
Prob (F-statistic)	0.540
Durbin-Watson	1.935

Coefficients

Variable	Coef	Std. Err	t	p-value
Intercept	4.61×10^5	6.93×10^5	0.666	0.506
year	-210.52	343.34	-0.613	0.540

- Regarding the **Bayesian regression**, we obtained that the mean of the posterior distribution of the coefficient corresponding to the years is -211.04 . This indicates that the population decreases per year.
- Furthermore, reviewing the Bayesian confidence intervals, it was found that there is a 94% probability that the year effect is between -822.54 and 444.45 . Thus, not statistically significant.

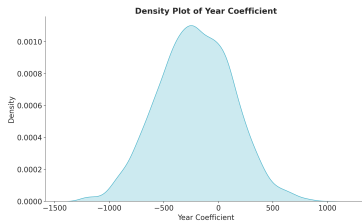
Bayesian Regression Results

Parameter	Mean	SD	94% HDI
Intercept	462101.05	690822.24	[-866169.01, 1689104.77]
year	-211.04	342.49	[-822.54, 444.45]
σ	40949.25	931.07	[39207.54, 42669.78]

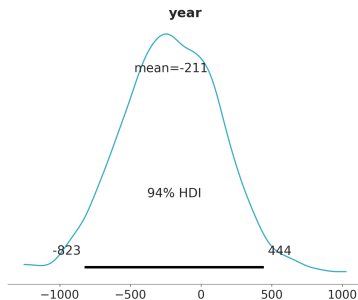
Effective sample size (ESS_{bulk}) $\approx$ 5354–6805, $\hat{R} = 1.00$ (good convergence).

- Regarding the MCMC chains, the chains appear to blend quite well, and the R-squared value is approximately 1, suggesting that the model converged correctly.

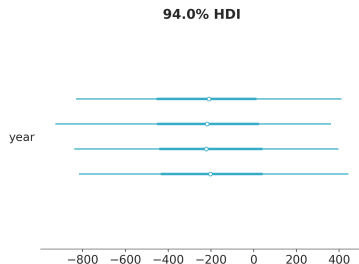
Densities



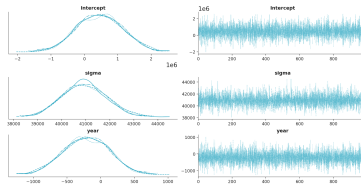
Confidence interval



Credibility Interval



Distribution



- Results did not show statistically significant changes over time.
- Due to this lack of significance, strong predictive conclusions cannot be made.
- No meaningful recommendations can be derived from the current model.

Conclusions

- Simple regression results were inconclusive.
- Population data are highly correlated with previous years.
- This violates the i.i.d. assumption of simple regression.
- Time series models may be more appropriate for this problem.